Fiscal Year:	FY 2024	**Task Last Updated:**	FY 07/04/2024
PI Name:	Massa, Gioia Ph.D.		
Project Title:	Pick-and-Eat Salad-Crop Productivity, Nutritional Value, and Acceptability to Supplement the ISS Food System		
Division Name:	Human Research, Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Food and Nutrition**:Risk of Performance Decrement and Crew Illness Due to Inadequate Food and Nutrition		
Space Biology Element:	(1) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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PI Organization Type:	NASA CENTER	**Phone:**	321-861-2938
Organization Name:	NASA Kennedy Space Center		
PI Address 1:	ISS Ground Processing and Research		
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PI Web Page:			
City:	Kennedy Space Center	**State:**	FL
Zip Code:	32899-0001	**Congressional District:**	8
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2013-14 HERO NNJ13ZSA002N-ILSRA. International Life Sciences Research Announcement
Start Date:	09/01/2015	**End Date:**	09/30/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Douglas, Grace	**Contact Phone:**	
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 9/30/2025 per HRP HHC element management (Ed., 8/10/21) NOTE: End date changed to 9/30/2021 per PI (Ed., 5/4/2020) NOTE: End date changed to 8/31/2020 per PI (Ed., 8/17/18) NOTE: Element change to Human Health Countermeasures; previously Space Human Factors & Habitability (Ed., 1/18/17) NOTE: Period of performance changed to 9/01/2015-8/31/2018 (previously 7/1/15-6/30/18) per G. Douglas/HRP (Ed., 4/3/16)		
Key Personnel Changes/Previous PI:	To better understand the microbiome and the transcriptome in the VEG-05 experiments we had to add additional collaborators to our team: Dr. Christina Khodadad, Dr. Anirudha Dixit, Cory Spern, and Dr. Natasha Haveman from Kennedy Space Center, and they contributed heavily to this report. Sydney Begerowski and Diana Arias from Johnson Space Center supported the behavioral health components of this study.		

COI Name (Institution):	Douglas, Grace Ph.D. (NASA Johnson Space Center) Hummerick, Mary M.S. (Amentum, Kennedy Space Center) Mitchell, Cary Ph.D. (Purdue University--grant NNX15AN78G) Morrow, Robert Ph.D. (Orbital Technologies Corporation) Wheeler, Raymond Ph.D. (NASA Kennedy Space Center) Young, Millennia Ph.D. (NASA Johnson Space Center) Spencer, LaShelle M.S. (Amentum, Kennedy Space Center) Buncheck, Jess M.S. (Southeastern Universities Research Association, Kennedy Space Center) Bell, Suzanne Ph.D. (NASA Johnson Space Center) Landon, Lauren (Wyle, Johnson Space Center)
Grant/Contract No.:	Internal Project
Performance Goal No.:	
Performance Goal Text:	
Task Description:	The capability to grow nutritious, palatable food for crew consumption during spaceflight has the potential to provide health promoting, bioavailable nutrients, enhance the dietary experience, and reduce launch mass as we move toward longer-duration missions. However, studies of edible produce during spaceflight have been limited, leaving a significant knowledge gap in the methods required to grow safe, acceptable, nutritious crops for consumption in microgravity. The "Veggie" vegetable-production system on the International Space Station (ISS) offers an opportunity to develop a "pick-and-eat" fresh vegetable component to the ISS food system as a first step to bioregenerative supplemental food production. We propose growing salad plants in the Veggie unit during spaceflight, focusing on the impact of light quality and fertilizer formulation on crop morphology, edible biomass yield, microbial food safety, organoleptic acceptability, nutritional value, and behavioral health benefits of the fresh produce. Phase A of the project would involve flight tests using leafy greens. Phase B would focus on dwarf tomato. Our work will help define light colors, levels, and horticultural best practices to achieve high yields of safe, nutritious leafy greens and tomatoes to supplement a space diet of prepackaged food. Our final deliverable will be the development of growth protocols for these crops in a spaceflight vegetable production system. Specific aim 1: Evaluate the effects of four light treatments and two different fertilizer compositions on the yield, morphology, organoleptic acceptability, and nutritional attributes of leafy greens during flight-definition and flight testing. Specific aim 2: Perform cultivar selection and evaluate the effects of four different red: blue light treatments and two different fertilizer compositions on the yield, morphology, organoleptic acceptability, and nutritional attributes of dwarf tomato during ground and flight tests. Specific aim 3: Perform hazard analysis, develop plans for minimizing microbial hazards, and screen flight-grown produce for potential pathogens.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Research Project: Our work on "Pick-and-Eat Salad-Crop Productivity, Nutritional Value, and Acceptability to Supplement the ISS Food System" focuses on developing a fresh food production capability on the International Space Station (ISS). We are using the Veggie hardware to develop light and fertilizer combinations that generate nutritious and appealing leafy green vegetables and dwarf tomatoes that astronauts can safely consume. The results of this research will directly translate to Earth-based controlled environment production of these and similar crops in vertical farms and urban plant factories. The capability to grow nutritious, palatable food for crew consumption during spaceflight can potentially provide health-promoting, bioavailable nutrients, enhance the dietary experience, and reduce launch mass as we move toward longer-duration exploration missions. However, studies of edible produce during spaceflight have been limited, leaving a significant knowledge gap in the methods required to grow safe, acceptable, nutritious crops for consumption in microgravity. The Veggie vegetable-production system on the ISS offers an opportunity to develop a "pick-and-eat" fresh vegetable component to the ISS food system as a first step to bioregenerative supplemental food production. Our goal is to grow salad crops in the Veggie unit during spaceflight and assess the impact of light quality and fertilizer formulation on crop morphology, edible biomass yield, microbial food safety, organoleptic acceptability, nutritional value, and behavioral health benefits. Our work will help define light color ratios, fertilizer composition, and horticultural best practices to achieve high yields of safe, nutritious leafy greens and tomatoes to supplement a space diet of prepackaged food. Our final deliverable will be to develop growth protocols for these crops in a spaceflight vegetable-production system. This will reduce the risk and close the gap of inadequate nutrition by helping us advance bioregenerative food production to supplement the packaged diet for future space exploration.
	Status Pick-and-eat Salad-crop Productivity, Nutritional Value, and Acceptability to Supplement the International Space Station (ISS) Food System (VEG-04A, VEG-04B, and VEG-05) is a set of hybrid experiments of plant research with human organoleptic and behavioral research. These experiments are sponsored by the Human Research Program but are implemented in partnership with the Space Biology Program. The VEG-04 flight experiments were conducted in 2019, and the VEG-05 experiment was conducted on ISS Dec. 2022 through March 2023. Because of the delay, an additional component was added to this experiment, known as HRF VEG (which stands for Human Research Facility – Veggie). The HRF VEG experiment consists of collecting data-of-opportunity from other space biology investigations with plants that were conducted on the ISS between the VEG-04 and VEG-05 tests. Data collected are specific to the human research foci of this project and include only behavioral health and performance data with the Veggie Questionnaire and Profile of Mood States data collected and Organoleptic acceptability data on the crops. I. Updates A. Student Augmentation of Grant A student augmentation grant titled "Pick-and-Eat Salad-Crop Nutrition & Next-Generation System Designs" was awarded by the Human Research Program to PhD student Jess Bunchek associated with this larger project. This augmentation enabled nutritional analysis of crop samples grown in the EDEN ISS plant production facility at the German Neumayer Station III in Antarctica and the aeroponic nutrient solutions with which the crops were grown. These data will provide a better understanding of crop nutrient composition across different plant

production platforms and across a long-duration expedition in an isolated, confined, extreme environment that has similarities to a space mission. With this augmentation, Jess was able to attend the HRP Investigators Workshop and present “Nutrition of Antarctic-Grown Crops to Supplement the Crew Diet, with Applications for Spaceflight.” A manuscript highlighting this experimentation is expected in FY24.
B. VEG-04A and B VEG-04A was conducted during Increment 57-58 and ran from June 4, 2019-July 9, 2019. VEG-04B was conducted during Increment 61-62 and ran from October 1, 2019-November 28, 2019, and both had ground controls run ~48 hours later. Science samples were returned from the ISS and were processed for microbiological food safety and chemical analyses.

An article, Pick-and-eat space crop production flight testing on the International Space Station, was published in Feb. 2024 in a special edition of The Journal of Plant Interactions, Article Collection on “Plant Astrobiology.” The citation is:

Jess M. Bunchek, Mary E. Hummerick, LaShelle E. Spencer, Matthew W. Romeyn, Millennia Young, Robert C. Morrow, Cary A. Mitchell, Grace L. Douglas, Raymond M. Wheeler & Gioia D. Massa (2024) Pick-and-eat space crop production flight testing on the International Space Station, Journal of Plant Interactions, 19:1, 2292220, DOI: 10.1080/17429145.2023.2292220 https:// and as of 6/25/2024 it has been viewed 1,234 times. This publication includes data on mizuna crop yield, chemistry, nutrient composition and microbial food safety, as well as human sensory data. The behavioral health data from VEG-04 A and B have been analyzed but we are still compiling these data with data in HRF-VEG and VEG-05 studies, and preliminary analyses are provided below.

C. HRF VEG All planned HRF VEG studies have been completed, and no additional tests have been added. Data are being compiled for publication as part of the BHP data set for this study.

D. VEG-05 Experiment Summary The Veggie (Vegetable Production System) on the ISS offers an opportunity to develop a “pick-and-eat” fresh vegetable component for the ISS food system as the first step towards bioregenerative supplemental food production. In the Veggie unit, during spaceflight, salad plants will be grown, focusing on the impact of light quality and fertilizer formulation on crop morphology, edible biomass yield, microbial food safety, organoleptic acceptability, nutritional content, and behavioral health benefits of the fresh produce for the Solanum lycopersicum "Red Robin" dwarf Tomato cultivar. The VEG-05 experiment will test different red-to-blue light ratios using the Veggie units on the ISS. Tomato plants will be grown in Veggie for ~100 days and astronaut crewmembers will provide plant care and pollination, and harvest fruit 3 times during and at the end of this growth period. Crewmembers will be asked to complete self-report surveys, including the Profile of Mood States – Short Form (POMS-SF) and a Veggie-specific questionnaire pre-flight, in-flight, and post-flight. They will also be asked to perform an organoleptic evaluation of a portion of the fresh produce they consume at harvest.

Consumed tomato samples will have their mass measured on ISS prior to crew consumption, and half of the produce will be frozen and returned to Earth for post-flight microbiological food safety and nutritional analysis. The goal of this experiment is to help define light colors and horticultural best practices to achieve high yields of safe, nutritious appealing tomato fruit to supplement a space diet of prepackaged food.

VEG-05 Flight Operations SpX-CRS-26 launched the VEG-05 payload on Nov. 26, 2022, and it docked on Nov. 27. Initiation of VEG-05 was started on 12/9/2022, but challenges led to subsequent initiation activities on 12/12/2022 and plant growth initiation on 12/14/2022, with the first ground control started 12/16/2022. This was later terminated due to poor germination and a second ground control was run 2/1/2023-05/12/2023. Both flight and ground control ran 100 days, with harvests of fruit at day 83, day 90, and day 100. In total, from the five surviving red-rich lighted plants in flight, only 5 ripe fruits were produced, and from the four surviving blue-rich lighted flight plants, 10 fruits were produced with only 6 of these ripe by day 100. A large number of photos and some video tours are part of the plant data collected. Due to the challenges with growth enumerated in the previous report, all planned activities were not accomplished. Human subject surveys were conducted but there were insufficient fruit samples for sensory evaluation. All fruit and plant samples were returned frozen at or below -80ºC. Plant samples included leaves and adventitious roots of surviving plants. Water samples were returned from root mat reservoirs, with the post-growth samples collected the day after harvest. Only a few swabs were taken from the flight Veggies from bungees only, due to harvest activities running over time. The planned samples from bellows and the Veggie fans were not collected. Pillows 1, 3, 7, and 10 were also collected and returned for analysis of roots, wicks, and substrates.

Because of the small sample sizes and factors affecting growth on the ISS, the objectives of assessing the light quality effect (red: blue light treatments) will not be fully achieved. Revised objectives of this study include comparing stressed flight plants with normal ground plants to determine the impact of plant overwatering stress in space on food safety and the plant microbial community, to determine nutrient content changes in fruit and leaves from stressed plants, and to evaluate stress metabolism changes in returned tissue by transcriptomic analysis. Postflight analysis was conducted with the following analyses being prioritized: A. culturable microbiology and food safety as well as molecular microbial community analysis of 1. ripe fruit; 2. leaves, stems, and adventitious roots; 3. pillow components (roots, wicks, and substrates); 4. swabs; and 5. water samples from root mats before and after growth. B. transcriptomics of leaf tissue and adventitious roots, and C. elemental analysis of leaf tissue. Unfortunately, there was insufficient tissue to conduct the analyses of leaves; however, elemental analyses were performed on a few of the harvested fruit.

Microbial Analyses Total Culturable Microorganisms were assessed with determinations of Aerobic Plate Counts (APC)/Bacteria and Total Yeast and Mold Count. Specific pathogens were screened for including Generic E. coli/Coliforms using Petrifilm, Salmonella sp. using enrichment/selective media, and Staphylococcus aureus using Petrifilm. Any colonies that grew in these methods were isolated and identified using the Biolog MicroID system and MicroSEQ fungal and bacterial gene sequencing ID. Screening for selected pathogens yielded negative results and this was confirmed by community 16S and ITS sequencing.

We observed no difference between red-rich and blue-rich treatments in bacterial and fungal CFUs in-flight samples. The red-rich ground control surface swabs and adventitious roots had significantly lower bacterial counts than the blue-rich treatment, while the fungal counts were lower in the red-rich wicks, leaves, and adventitious roots. Blue-rich fruit samples, surface swabs and leaves were lower in the ground controls than in the flight samples for both bacteria and fungi.

Molecular Analyses - Microbiome Community sequencing was performed using the 16S rRNA gene for bacterial taxonomic identification with the QIIME 2.0 with Silva approach. ITS was used for fungal taxonomic identification

Task Progress:

using KRAKEN2 with the UNITE database.

If we look at a heat map of the total microbiome, we see differences between flight and ground, and we see that true roots (from plant pillows) and adventitious roots had higher community diversity than leaf or fruit samples. When we look at 16S details by sample we see Pseudomonas dominating many of the samples, and if we remove that we start to see different clustering appear.

Molecular Analyses - Transcriptome Transcriptomics of leaf and adventitious roots sampled was performed with the GeneLab RNA-Seq Consensus Pipeline-RCP. We looked at the percentage distribution of differentially expressed genes (DEGs) in adventitious roots and leaves of ground and flight plants grown with either the red-rich or blue-rich lighting treatments and saw that leaf and root tissue in blue-rich lighting showed a higher percent of DEGs.

When Principal Component Analyses (PCA) are performed on these data, we start to see leaf and root DEGs clustered based upon condition (flight versus ground) and gene expression completely separated based on tissue type, but there is overlap in both tissue types between flight and ground. If PCA plots are coded to differentially represent red-rich and blue-rich treatments, there is considerable overlap between ground and flight samples in red-rich treatments, but blue-rich samples segregate between these locations.

Work is underway to interpret the differential gene expression observed in VEG-05. In leaves, there is strong differential regulation of genes in blue-rich grown samples in flight. Biological processes are identified, such as light-harvesting photosystem I, cell division, and protein regulation. There is a different group of upregulated genes in blue-rich grown ground control leaf samples with functions such as cell cycle regulation and protein biosynthesis being modified. In adventitious roots, we see some similar responses as in the leaf tissue including changes in photosynthesis, ATP energy production, and protein biosynthesis. This is not unexpected because the adventitious roots did contain some chlorophyll.

In the red-rich grown plant samples, there were fewer DEGs. For leaves, these involve respiration, mitochondria and energy production in the form of electron transport. In adventitious roots, DEGs are showing that these are responding to red-light differential regulation in photosynthesis and energy production, protein biosynthesis. When gene ontology is looked at, we start to see a broad categorization of the genes involved. For adventitious roots under the same conditions, we see similar responses to the leaves but also responses to stress, hydrogen peroxides catabolic processes or redox stress.

Overall, there appear to be more DEGs in leaf and adventitious root samples in blue-rich light compared to red-rich light. A functional enrichment analysis of DEGs showed similarities between leaf and adventitious root samples, as expected. Some of the hallmarks of spaceflight were detected in flight samples including stress response, lignin, radiation stress, and hydrogen peroxide catabolic processes. Further analyses are underway, and two manuscripts on the VEG-05 plant data are being prepared. While not generating the desired information on spaceflight growth responses of healthy crops, our team is hopeful that these analyses will shed light on tomato responses to stress in this environment as plant overwatering stress is a mission-relevant condition that could occur in future space crop growth systems.

Behavioral Health data from VEG-04, HRF VEG, and VEG-05 Plants as a Potential Spaceflight Behavioral Health and Performance (BHP) Countermeasure Plants are a potential countermeasure for the stresses of living in space. Caring for plants and eating fresh food have been shown to serve as a psychological benefit for previous astronauts and for others in analogous environments such as Antarctica (Schlacht et al., 2019, Vessel & Russo, 2015). Gardening can be therapeutic and has been used as such in terrestrial settings (Odeh & Guy, 2017). These effects may carry over or even be more pronounced in austere resource-limited environments such as spaceflight. It can reduce stress and increase sensory stimulation and enjoyment (Vessel & Russo, 2015). Growing crops for consumption may also offer astronauts a form of meaningful and engaging work as they care for living things in an austere environment away from Earth's nature and as they provide sustenance for their crew. Additionally, fresh produce may enhance the diet and encourage consistent consumption by avoiding menu fatigue, which may further support psychological health and well-being by supporting a healthy brain and body (Douglas et al., 2022). Despite these potential benefits, difficulty in crop production and struggling or dying plants may reduce positive effects. The data on the psychological effects of growing crops in spaceflight is limited.

As such, we examined behavioral outcomes associated with tending to plants and consuming crops grown during long-duration missions. Here, we further characterized which aspects of horticulture are most salient and psychologically beneficial to astronaut health and well-being. We hypothesized that engaging with plants during a mission would have a beneficial effect as evidenced by reported positive mood and enhanced well-being, meaningfulness, connection to Earth, relationships, and performance. We also hypothesized that interacting with the plants would be a source of positive sensory stimulation for the crew. Preliminary in-flight data analysis results are described below. More comprehensive analyses are underway in preparation for the final report.

Methods for the In-flight Phase for BHP Aims Participants Participants were 27 astronauts who interacted with the plant systems VEG-04, VEG-05, and HRF-VEG (Human Research Facility – Veggie). HRF-VEG data were collected from other space biology investigations of plants that were being conducted on the ISS between the VEG-04 and VEG-05 tests. HRF-VEG data included VEG-03 I, J, K, and L and PH-04. Total N sizes noted below vary by different plant activities, depending on what each astronaut may or may not have experienced during their mission. There was a total of 106 in-flight observations. Measures POMS-SF. The Profile of Mood States Short Form (Curran et al., 1995) was given to participants every three weeks around plant growth. Results forthcoming. BHP Veggie Questionnaire. The Behavioral Health and Performance Veggie Questionnaire was given to participants monthly, beginning with plant growth initiation. Plant and crop growth experience items examined enjoyment and time spent performing specific activities with the plants; behavioral health outcomes such as engagement, demanding, meaningful, and effects on well-being, mood, performance, and relationships with others; connections to the Earth; desire to work with and eat plants; sensory stimulation for sight, smell, touch, and taste; human factor aspects of time, training, and supplies; and experiences with struggling or dying plants. Items were initially offered on a 201 or 101-point visual analog scale (VAS), with higher ends of the scale indicating more positive outcomes, e.g., enhanced enjoyment, meaning, and engagement. The lower ends of the scale were anchored with more negative outcomes, e.g., diminished enjoyment, etc. The scale midpoints were neutral in that they neither enhanced nor diminished. Data was binned to a 7-point Likert scale at the direction of the study's biostatistician for use in our analyses. Participants selected a specific survey version based on whether they interacted with the plant system(s) or not, and we included only the scores from the crewmembers who participated in any given activity.

	Preliminary Results and Discussion for BHP Aims Generally, engaging in tasks related to the plants and crop growth (a.k.a., Veggie tasks), was enjoyable. The highest ratings were related to consuming the harvested plants and voluntary viewing, suggesting that some crewmembers found time to experience the plants and nature as a source of enjoyment. Tending to the plants was also moderately enjoyable (e.g., plant thinning, pollinating, wick opening, and harvesting), indicating that there are BHP benefits from this work task. The BHP impacts of whether the task was engaging, meaningful, supported well-being, or was demanding for a crewmember also indicates generally positive trends. Engaging, meaningful, and supporting well-being were strongly, and positively correlated with each other, which is consistent with behavioral health and workplace research that suggests that engaging and meaningful work can support well-being. Scores for these three positively-oriented variables increase over time, suggesting that the benefits of working with plants – that is, seeing them grow, tending to them, continuing to enjoy them as nature, and ultimately, consuming the fresh produce – becomes a greater source of behavioral health benefits as the long-duration mission continues. For plant activities, not only are they generally enjoyable for those crewmembers engaging with the plants, but the data show voluntary viewing, watering, harvesting, and consuming as consistently, positively related to the task being engaging and meaningful, and enhancing well-being. Preliminary analyses suggest that working with plants in space as well as consuming them, can be a behavioral health countermeasure for long-duration spaceflight. Additional analyses will be completed for the final report and a publication is in the works. References Curran, S. L., Andrykowski, M. A., & Studts, J. L. (1995). Short form of the profile of mood states (POMS-SF): psychometric information. Psychological assessment, 7(1), 80. Douglas, G. L., DeKerlegand, D., Dlouhy, H., Dumont-Leblond, N., Fields, E., Heer, M., ... & Zwart, S. R. (2022). Impact of diet on human nutrition, immune response, gut microbiome, and cognition in an isolated and confined mission environment. Scientific Reports, 12(1), 20847. Odeh, R., & Guy, C. L. (2017). Gardening for therapeutic people-plant interactions during long-duration space missions. Open Agriculture, 2(1), 1-13. Schlacht, I. L., Kolrep, H., Daniel, S., & Musso, G. (2020). Impact of plants in isolation: The EDEN-ISS human factors investigation in Antarctica. In Advances in Human Factors of Transportation: Proceedings of the AHFE 2019 International Conference on Human Factors in Transportation, July 24-28, 2019, Washington DC, USA 10 (pp. 794-806). Springer International Publishing. Vessel, E. A., & Russo, S. (2015). Effects of Reduced Sensory Stimulation and Assessment of Countermeasures for Sensory Stimulation Augmentation. A Report for NASA Behavioral Health and Performance Research: Sensory Stimulation Augmentation Tools for Long Duration Spaceflight (NASA/TM-2015-218576). NASA Johnson Space Center, Houston, TX.
Bibliography Type:	Description: (Last Updated: 06/24/2026)
Articles in Peer-reviewed Journals	Bunchek JM, Hummerick ME, Spencer LE, Romeyn MW, Young M, Morrow RC, Mitchell CA, Douglas GL, Wheeler RM, Massa GD. "Pick-and-eat space crop production flight testing on the International Space Station." J Plant Interact. 2024 Jan 18;19(1):2292220. https://doi.org/10.1080/17429145.2023.2292220 , Jan-2024
Articles in Peer-reviewed Journals	Boles HO, Poulet L, Johnson CM, Torres JJ, Koss LL, Spencer LE, Massa GD. "Design, build, and testing of hardware to safely harvest microgreens in microgravity." Gravit Space Res. 2023 Oct 30;11(1):1-14. https://doi.org/10.2478/gsr-2023-0001 , Oct-2023